CHAPTER 01

1. Introduction

1.1 Domain Background

Alzheimer's disease is a degenerative neurological condition that impairs memory, reasoning, and cognitive abilities. It is the leading cause of dementia, with almost 70% of worldwide cases. Initially, patients have minor memory loss, like forgetting names, appointments, or daily activities, but are capable of independent living. As the condition develops, problems in orientation, communication, and making decisions escalate, causing confusion and lower confidence. The World Health Organization (WHO) states that over 55 million people around the globe now suffer from dementia, and the number is anticipated to grow at a high rate in the next few years.

The effects of Alzheimer's are not only felt by patients but also by families and caregivers, who endure great emotional and physical burdens. The anxiety and frustration experienced by early-stage patients stem from their knowledge of cognitive loss. Thus, care technologies like reminder systems, facial recognition, and voice-enabled assistance can promote independence, mitigate caregiver stress, and enhance overall quality of life.

1.2 Motivation and Significance of the Project

The project is motivated by the growing difficulty experienced by persons who are in the initial stages of Alzheimer's disease and their caregivers. The patients may have trouble with memory loss, getting lost, and performing daily activities, while the caregivers go through immense emotional and physical stress. Reminder or support software available is not formulated keeping cognitive impairments in view, which restricts its efficiency This project will create an intelligent, user-friendly assistive technology that promotes memory and autonomy through the use of voice commands, facial recognition, and smart reminders.

The plan will improve emotional health, ease caregiver stress, and improve everyday functioning. The creation of technologically assisted dementia care, as prioritized by the World Health Organization's highest global health priorities, is what makes it important (WHO, 2023).

1.3 Objectives and Scope of the Project

The primary objective of this project is to develop a mobile assistive application that allows people with early-stage Alzheimer's disease to manage daily tasks on their own. The primary objectives are:

- Using both audios to remind people to take their medications, eat, and attend appointments.

- Assisting with face recognition to identify family members or frequent guests.

- Using voice-based communication to help patients who are unsure or confused.

- Giving caregivers a way to upload images, issue reminders remotely, and monitor patient activities through an attached dashboard.

The project's scope is limited to people suffering from early-stage Alzheimer's disease who can use mobile devices with little guidance. Advanced stages of the illness that call for medical or physical support won't be handled by the system.

1.4 Proposed Solution

According to this project, a mobile application designed for Alzheimer's patients should be developed. The application will have four main modules:

- **Reminder System** – to remind patients about medication timing, appointments, and key activities.

- **Face Recognition** – to enable patients to recognize familiar faces, including relatives and regular visitors, through a lightweight machine learning model.

- **Voice Assistance** – provides voice reminders to patients for medications and activities, reducing the need for manual interaction.

- **Caregiver Dashboard** – enabling caregivers to input reminders, add visitor photos, and track logs of tasks accomplished, thus simplifying their work.

In integrating these, the system caters to memory loss, difficulty in recognition, and dependency, providing a solid support system for both caregivers and patients.

1.5 Expected Outcomes

The outcome of this project is a usable and understandable mobile application that improves the standard of living for Alzheimer's patients. The patients will be assisted in remembering things to do and identifying their dear ones, thus alleviating the anxiety and feeling of loneliness. The caregivers will be helped by decreased workload and better monitoring facilities, enabling them to manage their duties more efficiently. Ultimately, the project seeks to show how technology can become an integral part of Alzheimer's care by enabling patients, mitigating caregiver stress, and facilitating more dignified and independent living for those suffering from the disease.

1.6 Main Contributions

- Development of a mobile assistive application tailored for early-stage Alzheimer's patients.
- Integration of reminder scheduling, face recognition, and voice-based assistance in one platform.
- Implementation of a light-weight deep learning model for real-time face recognition.
- Creation of a caregiver dashboard for remote monitoring and management.
- Performance evaluation based on usability, accuracy, and patient-caregiver satisfaction.

1.7 Problem Statement

Current digital assistants like medication reminder software, health monitoring devices, and overall assistive devices support at most to some extent but are not tailored for use by individuals suffering from Alzheimer's disease. Most of them do not have cognitive impairment management capabilities, for instance, intelligent voice control to assist confused patients or face recognition to recognize known individuals. These patients become more and more dependent on their caregivers because they are consistently missing important appointments. They keep forgetting to turn in their medications or can't recognize even familiar faces. It reinforces the importance of using a full, patient-centered support system that can accommodate the specific behavioral changes and mental limitations of patients with Alzheimer's disease. This concern would alleviate some of the burden on caregivers as well as have a significant positive impact on the autonomy, safety, and quality of life for the patients.

1.8 Proposed Solution

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In integrating these, the system caters to memory loss, difficulty in recognition, and dependency, providing a solid support system for both caregivers and patients.

1.9 Main Objectives

- To acquire and preprocess a dataset of caregiver-Alzheimer's patient interactions to train and test system modules.

- To create a mobile-based assistive system with reminder scheduling, face recognition, and voice-based interaction.

- To develop and deploy a light-weight deep learning model for real-time face recognition of known individuals.

- To integrate an intelligent voice assistant with the ability to offer verbal prompts, reminders, and visitor identification.

- To create a caregiver dashboard for remote monitoring of reminders, visitor information, and patient activity logs.

- To measure system performance against usability, accuracy, and patient-caregiver satisfaction measures.

1.10 Assumptions & Constraints

- The project's scope is limited to people suffering from early-stage Alzheimer's disease who can use mobile devices with little guidance.

- Advanced stages of the illness that call for medical or physical support won't be handled by the system.

- The incremental structure will lower risk and provide users with a range of options when changes are necessary.

- This iterative structure guarantees that risks are kept to a minimum.

- The system can continue to adjust to changing user requirements.

1.11 Project Scope

The project's scope is limited to people suffering from early-stage Alzheimer's disease who can use mobile devices with little guidance. Advanced stages of the illness that call for medical or physical support won't be handled by the system.

1.12 Software Development Lifecycle Model

For this project, the Incremental Development Model will be applied. Instead of deploying a whole system, this paradigm enables the system to be constructed and tested in tiny, manageable portions. Voice assistance,

facial recognition, and reminder alerts are just a few of the helpful features that are included in every iteration for the user.

The following are the primary phases of this methodology:

- Requirement Analysis: Involves talking with medical professionals and caregivers about what the possible system would look like.

- System Design: building designs at both the architectural level and component level, as shown in Figure 1.

- Execution and Confirmation: Modifying modules incrementally with testing of each module, sequentially.

- Integration and Verification: Run the end-to-end verification test . Make the modules work together.

The incremental structure will lower risk and provide users with a range of options when changes are necessary.

This iterative structure guarantees that risks are kept to a minimum. The system can continue to adjust to changing user requirements.

CHAPTER 02

2 Requirements Analysis

Requirements analysis is the process of identifying, documenting, and validating the functional and non-functional requirements of a system. It defines what the system must do and under what constraints it must operate.

For the *ForgetMeNot* system, requirements are derived from:

- Alzheimer's patient needs (early-stage)
- Caregiver operational needs
- System safety and usability expectations
- AI-based functional constraints (face recognition & voice)

2.1 Literature Review / Existing System Study

Numerous technology solutions have been put out in recent years to help people with dementia and Alzheimer's disease. Reminders, facial recognition, wearable support, and communication aids are the main functions of these systems. Some of the significant existing systems and studies are discussed below.

1. C. Fang et al.

C. Fang et al. [1] conducted a clinical study to examine the impairment in holistic facial recognition processing among Alzheimer's patients. Understanding how Alzheimer's disease impacts the brain's capacity to interpret and identify familiar faces is the main goal of their research. According to the study, patients frequently struggle to identify close relatives because of cognitive processing issues with facial features in addition to memory loss. The results show that the problem goes beyond simple forgetfulness and offer a neurobiological explanation for facial recognition problems.

Because it pinpoints the underlying cause of recognition impairment and demonstrates the necessity of technology support systems, this study is noteworthy. But the study is only theoretical and clinical in character. No assistive technology is suggested or used to help patients deal with this difficulty in everyday life. Therefore, it lacks a workable answer for everyday application even though it offers solid core information.

2. Akare et al.

U. Akare et al. [2] proposed a laptop-based assistive system that integrates facial recognition with medication reminder functionality. The system uses Convolutional Neural Networks to detect and recognize faces through a webcam. Once a face is identified the system audibly announces the person's name to assist the patient in recalling their identity. In addition to facial recognition the system also provides scheduled medication reminders to improve treatment adherence.

The integration of recognition and reminder features makes this system more comprehensive than single purpose solutions. However the system is limited by its dependency on a laptop which reduces portability and practical usability in everyday environments. It also lacks advanced caregiver monitoring tools and

mobile accessibility which are important for real time supervision and convenience. As a result, the system demonstrates feasibility but remains limited in scalability and real world application.

3. Anshi Paul

Anshi Paul [3] developed a wearable facial recognition device specifically designed for dementia patients. The device captures images in real time and provides voice feedback by identifying familiar individuals. This functionality helps patients recognize people during social interactions reducing confusion, anxiety and social withdrawal. The wearable nature of the device increases mobility and allows patients to receive assistance without being restricted to a fixed location.

Although the system improves independence and social confidence it remains at the prototype stage with limited large scale validation. The device focuses primarily on facial recognition and does not integrate additional supportive features such as medication reminders , conversational AI or caregiver dashboards. Therefore while it addresses recognition challenges effectively it does not provide a complete assistive solution as the system is still at the prototype stage.

4. Peres & Campos

Peres and Campos [4] conducted a systematic review of reminder and guidance technologies developed for dementia patients.Their study compares several reminder systems and finds that usability, customisation, simplicity, and caregiver involvement are important characteristics that affect how effective they are. The research highlights that for assistive technologies to produce significant results they must be user friendly and customized to meet the needs of each patient.

The study emphasizes the significance of human centered development in assistive systems and offers insightful design recommendations. It does not, however, outline the creation or application of a particular system. Rather, it provides critical perspectives on current technologies. Although the study advances knowledge of best practices, it falls short of offering a cohesive, workable solution that integrates several assistive elements into a single platform.

5. Memoro (MIT Media Lab)

The Memoro Project [5] is an AI-powered memory assistance system designed to support individuals suffering from dementia and Alzheimer’s disease. It combines a wearable device with Personal AI to help patients store, retrieve, and recall important memories. Unlike traditional reminder systems that provide fixed time-based alerts, Memoro focuses on contextual memory support. The system attempts to analyze interactions and situations in order to provide relevant prompts, such as reminding patients about people they meet, past events, or important personal information.

In contrast to straightforward notification-based solutions, the project signifies a move toward intelligent and adaptable memory support. However privacy and security issues become major obstacles because it mostly depends on gathering and processing personal data. Furthermore proper contextual awareness and ongoing learning are essential to the system's efficacy. Memoro offers a novel AI based strategy but it

falls short in combining facial recognition and reminder capabilities into a single, easily navigable platform, simplifying mobile interfaces, and providing organized caregiver management tools.

6. RAZ Memory Cell Phone

The RAZ Memory Cell Phone [6] is a commercially available device made especially for elderly and Alzheimer's sufferers. To lessen confusion and cognitive stress it offers a streamlined interface with big buttons and few alternatives. To make sure the patient can only call people they have permission to call caregivers can remotely manage contacts and settings. The gadget makes communication easier for people with memory impairments by emphasizing safety and use.

Although the RAZ Memory Phone is practical and user friendly it lacks advanced technological capabilities. It does not incorporate artificial intelligence, facial recognition, adaptive reminders or conversational support. The system mainly simplifies calling functionality rather than offering comprehensive cognitive assistance. Therefore, while it is effective as a communication aid it does not provide an integrated AI-driven solution for broader memory and recognition challenges.

Table 1: Comparison of Existing Systems & Proposed Solution.

System / Study	Type	Focus	Technology	Users	Strengths	Limitations
C. Fang et al. [1]	Clinical study	Facial recognition impairment	Behavioral tests	Alzheimer's	Explains recognition issues	No assistive system
Akare et al. [2]	Laptop system	Face ID + medication reminders	CNN, webcam, audio	Alzheimer's	Combines recognition & reminders	Laptop-based, no caregiver link
Anshi Paul [3]	Wearable device	Real-time face ID	Camera, voice feedback	Dementia / Alzheimer's	Portable, aids interaction	Prototype, small-scale
Peres & Campos [4]	Review	Reminder systems overview	Literature analysis	Dementia / Alzheimer's	Highlights usability factors	No implementation
Memoro [5]	Wearable AI	Memory recall & prompts	Personal AI, ML	Dementia / Alzheimer's	Context-aware memory aid	Privacy & validation limits
RAZ Memory Phone [6]	Commercial	Simplified memory aid	Phone + caregiver setup	Seniors / Alzheimer's	Easy to use, proven	No AI or adaptivity

Proposed System	Mobile app	Face ID + voice reminders + caregiver tools	CNN, TTS, dashboard	Alzheimer's & caregivers	Fully integrated, hands-free	Needs testing & privacy control

2.2 Stakeholders (Actors)

2.2.1 Primary Stakeholders

1. **Alzheimer's Patient**
    a. Uses the mobile app
    b. Receives reminders
    c. Uses voice assistance
    d. Gets visitor identification support
2. **Caregiver**
    a. Configures reminders
    b. Uploads visitor images
    c. Monitors logs
    d. Manages patient data

2.3 Requirements Elicitation

Requirements are divided into:

- Functional Requirements (FR)
- Non-Functional Requirements (NFR)

2.3.1 Functional Requirements

1. Caregiver Registration

1.1 Caregiver Registration Screen

Req ID	Requirement	Priority
FR-REG-01	The system shall provide fields for Full Name, Email, Password, and Confirm Password.	High
FR-REG-02	The system shall validate proper email format before submission.	High
FR-REG-03	The system shall ensure the password is at least 8 characters long.	High
FR-REG-04	The system shall ensure Password and Confirm Password match.	High
FR-REG-05	The system shall prevent registration if any required field is empty.	High
FR-REG-06	If an email already exists, the system shall display: “This email is already registered.”	High
FR-REG-07	The system shall create caregiver accounts using Firebase Authentication.	High

FR-REG-08	The system shall send an email verification link after successful registration.	
FR-REG-09	The caregiver shall not access the dashboard until email is verified.	High

2. Caregiver Login

Req ID	Requirement	Priority
FR-LOGIN-01	The login screen shall contain Email and Password fields.	High
FR-LOGIN-02	The system shall authenticate credentials using Firebase Authentication.	High
FR-LOGIN-03	If credentials are invalid, the system shall display an error message.	High
FR-LOGIN-04	If email is not verified, the system shall prompt a verification message.	
FR-LOGIN-05	Upon successful login, the caregiver shall be redirected to Dashboard.	High
FR-LOGIN-06	The system shall maintain a session unless the user logs out manually.	Medium

3. Patient Profile Setup & Linking

3.1 Initial Patient Setup

Req ID	Requirement	Priority
FR-LINK-01	After first login, the caregiver shall create a Patient Profile.	High
FR-LINK-02	The patient profile shall include Name, Age, Relationship, and optional photo.	High
FR-LINK-03	The system shall generate a unique Patient ID.	High
FR-LINK-04	All reminders, visitors, and logs shall be linked to Patient ID.	High

3.2 Device Pairing (Patient Mode Activation)

Req ID	Requirement	Priority
FR-LINK-05	Caregiver shall log in once on the patient's device to activate Patient Mode.	High
FR-LINK-06	After pairing, the device shall switch automatically to Patient Mode.	High

FR-LINK-07	Patient Mode shall not require daily login.	High
FR-LINK-08	Accessing caregiver controls on patient devices shall require password re-authentication.	High

3.3 Multiple Patient Support (Secondary Feature)

Req ID	Requirement	Priority
FR-LINK-09	The system shall allow one caregiver to manage multiple patients.	Medium
FR-LINK-10	Each patient shall have a unique Patient ID.	High
FR-LINK-11	Caregivers shall be able to switch between patients from Dashboard.	Medium
FR-LINK-12	Data of each patient shall remain isolated.	High

4. Caregiver Dashboard

Req ID	Requirement	Priority

FR-DASH-01	Dashboard shall display selected patient name and profile photo.	High
FR-DASH-02	Dashboard shall display summary of Today's reminders (Total, Completed, Missed).	High
FR-DASH-03	Dashboard shall provide navigation to Reminders, Visitors, Logs, Settings.	High
FR-DASH-04	If multiple patients exist, the dashboard shall indicate an active patient profile.	Medium

5. Reminder Management

5.1 Add Reminder (Caregiver)

Req ID	Requirement	Priority
FR-REM-01	Caregivers shall create reminders with Title, Type, Date, and Time.	High
FR-REM-02	Caregiver shall select repeat options (One-time, Daily, Weekly, Custom).	High
FR-REM-03	Caregiver shall add a custom voice message for reminder.	High
FR-REM-04	The system shall validate mandatory fields before saving.	High

FR-REM-05	Reminders shall be stored against the selected Patient ID.	High

5.2 Reminder Execution (Patient Side)

Req ID	Requirement	Priority
FR-REM-06	At scheduled time, the system shall display a full-screen alert.	High
FR-REM-07	The system shall automatically speak a reminder message using TTS.	High
FR-REM-08	Patients shall have options: “Done” and “Remind me in 5 minutes”.	High
FR-REM-09	If no response in 5 minutes, the reminder shall repeat once.	Medium
FR-REM-10	If no response after the second alert, the reminder shall be marked Missed.	
FR-REM-11	Reminder status shall be logged automatically.	High
FR-REM-12	Reminders shall function offline using local storage.	Medium

6. Visitor Management (Caregiver)

Req ID	Requirement	Priority
FR-FACE-01	Caregiver shall add Visitor Name and Relationship.	High
FR-FACE-02	Caregiver shall upload a minimum 1 photo (recommended 3–5).	High
FR-FACE-03	Photos may be captured via camera or selected from the gallery.	High
FR-FACE-04	The system shall generate face embeddings for recognition.	High
FR-FACE-05	Visitor data shall be linked to Patient ID.	High

7. Automatic Face Recognition (Patient Side)

7.1 Auto-Recognition Logic

Req ID	Requirement	Priority
FR-FACE-AUTO-01	The system shall support Automatic Face Recognition Mode.	High

FR-FACE-AUTO-02	When enabled, the system shall detect the human face without requiring button press.	High
FR-FACE-AUTO-03	Upon detection, the system shall capture and process images automatically.	High
FR-FACE-AUTO-04	If the confidence ≥ threshold, the system shall announce visitor name and relationship.	High
FR-FACE-AUTO-05	If there is no match, the system shall announce “I could not recognize this person.”	High
FR-FACE-AUTO-06	Recognition shall complete within 5 seconds.	Medium
FR-FACE-AUTO-07	Each recognition event shall be logged.	Medium

7.2 Cooldown Mechanism

Req ID	Requirement	Priority
FR-FACE-AUTO-08	The system shall not re-trigger recognition for the same face within 60 seconds.	Medium

7.3 Auto-Recognition Control (Caregiver)

Req ID	Requirement	Priority
FR-FACE-AUTO-09	Caregiver shall enable or disable Auto-Recognition Mode from Settings.	High
FR-FACE-AUTO-10	If disabled, face recognition shall only activate manually.	Medium

8. Voice Assistant

Req ID	Requirement	Priority
FR-VOICE-01	All reminders shall be spoken automatically using TTS.	High
FR-VOICE-02	Face recognition results shall be announced via voice.	High
FR-VOICE-03	The patient shall activate voice input using the “I Need Help” button.	High
FR-VOICE-04	The system shall support commands: “What are my reminders?”	High
FR-VOICE-05	The system shall support the command: “Call my caregiver.”	Medium

FR-VOICE-06	The system shall respond verbally to recognized commands.	High

9. Language Selection

Req ID	Requirement	Priority
FR-LANG-01	Caregiver shall select application language (English or Urdu).	Medium
FR-LANG-02	Reminder announcements shall follow selected language.	High
FR-LANG-03	Face recognition announcements shall follow selected language.	High

10. Activity Logs

Req ID	Requirement	Priority
FR-LOG-01	The system shall log reminders triggered, completed, and missed events.	High
FR-LOG-02	The system shall log face recognition events and results.	High

FR-LOG-03	Caregiver shall filter logs by date.	Medium
FR-LOG-04	Logs shall be stored securely in Firebase Firestore.	High

2.3.2 Non-Functional Requirements

NFR01 – Usability

Req. ID	Non-Functional Requirement
NFR01-01	The system interface shall be simple and cognitively friendly.
NFR01-02	The patient interface shall minimize text-based navigation.
NFR01-03	Large buttons and high-contrast colors shall be used.

NFR02 – Performance

Req. ID	Non-Functional Requirement

NFR02-01	Reminder delivery latency shall not exceed 3 seconds.
NFR02-02	Face recognition response time shall be under 10 seconds.
NFR02-03	System shall operate smoothly on mid-range Android devices.

NFR03 – Security

Req. ID	Non-Functional Requirement
NFR03-01	All user authentication shall use secure authentication.
NFR03-02	Patient data shall be encrypted in cloud storage.
NFR03-03	Unauthorized users shall not access patient records.

NFR04 – Privacy

Req. ID	Non-Functional Requirement

NFR04-01	Visitor images shall only be accessible to authorized caregivers.
NFR04-02	The system shall comply with ethical data handling practices.

NFR05 – Reliability

Req. ID	Non-Functional Requirement
NFR05-01	The system shall operate continuously without crash during daily use.
NFR05-02	Reminder scheduling shall not fail under normal operation.

NFR06 – Availability

Req. ID	Non-Functional Requirement
NFR06-01	The system shall be accessible 24/7.
NFR06-02	Cloud databases shall ensure high uptime.

NFR07 – Maintainability

Req. ID	Non-Functional Requirement
NFR07-01	The layered architecture shall allow independent module updates.
NFR07-02	The AI model shall be replaceable without affecting the UI layer.

2.4 USE CASE DESCRIPTIONS

2.4.1 UCD001 – Caregiver Registration

Use Case ID: 001	**Use Case Name: Caregiver Registration**
Priority: High	
Actors: Caregiver	
Use Case Summary:	The caregiver registers an account by providing full name, email, and password. The system validates all fields, creates an account via Firebase Authentication, and sends a verification email. The caregiver cannot access the dashboard until the email is verified.
Pre-Condition:	The caregiver must have a valid email address not previously registered in the system.

Normal Flow of Events	**Alternative Path**
1. Caregiver opens the application and navigates to the Registration screen.	Action cancelled.
2. Caregiver enters Full Name, Email, Password, and Confirm Password.	If any field is empty or invalid, the system displays an error message.
3. System validates email format and password strength (minimum 8 characters).	If an email already exists, the system displays: "This email is already registered."

4. The system checks if email already exists in the database.	
5. Caregiver clicks the "Register" button.	
6. The system creates an account using Firebase Authentication.	
7. The system sends a verification email to the provided address.	
8. The system prompts caregivers to verify their email before proceeding.	
Post Condition:	Caregiver account is successfully created. The caregiver must verify their email before accessing the dashboard.

2.4.2 UCD002 – Caregiver Login

Use Case ID: 002	**Use Case Name: Caregiver Login**
Priority: High	
Actors: Caregiver	
Use Case Summary:	The caregiver logs into the application using registered email and password. The system authenticates credentials via Firebase and redirects to the dashboard on success.
Pre-Condition:	The caregiver must have a registered and email-verified account.
Normal Flow of Events	**Alternative Path**
1. Caregiver opens the application and navigates to the Login screen.	Action cancelled.
2. Caregiver enters Email and Password.	
3. Caregiver clicks the "Login" button.	
4. The system authenticates credentials using Firebase Authentication.	If credentials are invalid, the system displays an error message.
5. The system checks whether email is verified.	
6. Upon success, the caregiver is redirected to the Dashboard.	

7. The system maintains sessions until manual logout.	
Post Condition:	Caregiver is logged in and directed to the Dashboard. Session is maintained until the caregiver logs out manually.

2.4.3 UCD003 – Patient Profile Setup

Use Case ID: 003	**Use Case Name: Patient Profile Setup**
Priority: High	
Actors: Caregiver	
Use Case Summary:	After first login, the caregiver creates a patient profile with name, age, relationship, and an optional photo. The system generates a unique Patient ID and links all future data to this profile.
Pre-Condition:	The caregiver must be logged in and no patient profile must exist yet.
Normal Flow of Events	**Alternative Path**
1. Caregiver logs in and is prompted to create a patient profile.	Action cancelled.
2. Caregiver enters Patient Name, Age, and Relationship.	
3. Caregiver optionally uploads a patient photo.	Skip photos if not available.
4. Caregiver clicks "Create Profile."	If mandatory fields are empty, the system displays an error.
5. The system generates a unique Patient ID.	
6. The system stores the profile and links it to the caregiver's account.	
7. Caregiver is redirected to the Dashboard.	
Post Condition:	The patient profile is created with a unique Patient ID. All subsequent reminders, visitors, and logs are linked to this profile.

2.4.4 UCD004 – Device Pairing (Patient Mode Activation)

Use Case ID: 004	**Use Case Name: Device Pairing**
Priority: High	
Actors: Caregiver	
Use Case Summary:	The caregiver logs in once on the patient's device to activate Patient Mode. The device then switches automatically to Patient Mode and no longer requires daily login.
Pre-Condition:	Caregivers must have a registered account and patient profile. The application must be installed on the patient's device.

Normal Flow of Events	**Alternative Path**
1. Caregiver opens the application on the patient's device.	If credentials are invalid, the system displays an error.
2. Caregiver logs in using their credentials.	Press cancel to abort.
3. The system detects the pairing intent and prompts confirmation.	
4. Caregiver confirms pairing.	
5. The system activates Patient Mode on the device.	
6. Device switches automatically to the Patient Mode interface.	
7. Accessing caregiver controls on this device now requires password re-authentication.	
Post Condition:	The device is successfully paired and running in Patient Mode. The patient does not need to log in daily.

2.4.5 UCD005 – Add Reminder

Use Case ID: 005	**Use Case Name: Add Reminder**
Priority: High	
Actors: Caregiver	

Use Case Summary:	The caregiver creates a reminder by specifying title, type, date, time, repeat frequency, and an optional custom voice message. The reminder is stored against the linked Patient ID.
Pre-Condition:	Caregivers must be logged in and a patient profile must exist.

Normal Flow of Events	**Alternative Path**
1. Caregiver navigates to the Reminders section from the Dashboard.	If no photo is provided, the system displays a validation error.
2. Caregiver clicks "Add Reminder."	If mandatory fields are missing, system displays a validation error.
3. Caregiver enters Title, Type, Date, and Time.	
4. Caregiver selects repeat options: One-time, Daily, Weekly, or Custom.	
5. Caregiver optionally records or types a custom voice message.	
6. Caregiver clicks "Save."	
7. The system stores the reminder linked to the Patient ID.	

Post Condition:	The reminder is saved and will trigger at the scheduled time on the patient's device.

2.4.6 UCD006 – Reminder Execution (Patient Side)

Use Case ID: 007	**Use Case Name: Reminder Execution (Patient Side)**
Priority: High	
Actors: Alzheimer's Patient	
Use Case Summary:	At the scheduled time, the system triggers a full-screen alert and automatically speaks the reminder message using TTS. The patient can mark it done or snooze for 5 minutes.

Pre-Condition:	The patient's device must be in Patient Mode with a reminder scheduled for the current time.

Normal Flow of Events	**Alternative Path**
1. System detects the scheduled reminder time.	The patient clicks "Remind me in 5 minutes" to snooze.
2. The system displays a full-screen alert with the reminder message.	If there is no response in 5 minutes, the system repeats the alert once.
3. The system automatically speaks the message via TTS.	If no response after the second alert, the reminder is marked Missed and logged.
4. The patient clicks "Done" to mark the reminder as Completed.	The patient clicks "Remind me in 5 minutes" to snooze.
5. System logs the reminder as Completed.	

Post Condition:	Reminder is marked as Completed or Missed and automatically logged for caregiver review.

2.4.7 UCD007 – Add Visitor (Face Registration)

Use Case ID: 007	**Use Case Name: Add Visitor (Face Registration)**
Priority: High	
Actors: Caregiver	
Use Case Summary:	The caregiver registers a visitor by providing their name, relationship, and at least one photo. The system generates face embeddings and links the visitor data to the Patient ID.
Pre-Condition:	Caregivers must be logged in and a patient profile must exist.

Normal Flow of Events	**Alternative Path**
1. Caregiver navigates to the Visitors section from the Dashboard.	If no photo is provided, the system displays a validation error.
2. Caregiver clicks "Add Visitor."	Press cancel to discard.

3. Caregiver enters Visitor Name and Relationship.	
4. Caregiver uploads at least 1 photo (recommended 3–5) via camera or gallery.	
5. Caregiver clicks "Save."	
6. The system generates face embeddings from the uploaded photos.	
7. The system stores visitor data linked to the Patient ID.	
Post Condition:	The visitor is successfully registered. The patient's device can now recognize this visitor during automatic face recognition.

2.4.8 UCD008 – Automatic Face Recognition (Patient Side)

Use Case ID: 008	**Use Case Name: Automatic Face Recognition**
Priority: High	
Actors: Alzheimer's Patient	
Use Case Summary:	With Auto-Recognition Mode enabled, the system continuously monitors the camera feed. Upon detecting a face, it captures and processes the image and announces the visitor's identity via voice.
Pre-Condition:	The patient's device must be in Patient Mode with Auto-Recognition Mode enabled and visitor data must be registered.

Normal Flow of Events	**Alternative Path**
1. Patient Mode is active and Auto-Recognition Mode is enabled.	If no match is found, the system announces: "I could not recognize this person."
2. The system detects a human face in the camera feed.	
3. The system automatically captures and processes the image.	

4. The system compares faces against registered visitor embeddings.	
5. If confidence meets threshold, the system announces visitor name and relationship via voice.	
6. The recognition event is automatically logged.	
7. The system enforces a 60-second cooldown before re-triggering for the same face.	
Post Condition:	The patient is verbally informed of the visitor's identity. Recognition event is logged for caregiver review.

2.4.9 UCD009 – Voice Assistance (Patient Side)

Use Case ID: 009	**Use Case Name: Voice Assistance (Patient Side)**
Priority: High	
Actors: Alzheimer's Patient	
Use Case Summary:	The patient presses the "I Need Help" button to activate voice assistance. The system listens to voice commands, processes them, and responds verbally with the requested information.
Pre-Condition:	The patient's device must be in Patient Mode with microphone access granted.
Normal Flow of Events	**Alternative Path**
1. The patient presses the "I Need Help" button on the screen.	If the command is unrecognized, the system asks the patient to repeat.
2. The system activates the microphone and listens for a voice command.	
3. The patient speaks a command, e.g., "What are my reminders?"	

4. The system processes the command and fetches relevant data.	
5. The system verbally responds to the patient with the requested information.	
6. If a patient says "Call my caregiver," the system initiates a call.	
Post Condition:	The patient receives verbal assistance and their need is addressed.

2.4.10 UCD010 – Language Selection

Use Case ID: 010	**Use Case Name: Language Selection**
Priority: Medium	
Actors: Caregiver	
Use Case Summary:	The caregiver selects the preferred application language (English or Urdu) from Settings. All reminder and face recognition announcements are then delivered in the selected language.
Pre-Condition:	Caregivers must be logged in with access to the Settings section.

Normal Flow of Events	**Alternative Path**
1. Caregiver navigates to Settings from the Dashboard.	Press cancel to discard changes.
2. Caregiver selects preferred language: English or Urdu.	
3. Caregiver clicks "Save."	
4. System updates the language setting for the linked patient profile.	
5. All subsequent TTS announcements follow the selected language.	

Post Condition:	Language preference is saved. All announcements are delivered in the selected language.

2.4.11 UCD011 – View Activity Logs

Use Case ID: 011	**Use Case Name: View Activity Logs**	
Priority: High		
Actors: Caregiver		
Use Case Summary:	The caregiver views a history of reminder and face recognition events for the linked patient. Logs can be filtered by date for focused review.	
Pre-Condition:	Caregivers must be logged in and at least one log entry must exist.	
Normal Flow of Events		**Alternative Path**
1. Caregiver navigates to the Logs section from the Dashboard.		Skip filtering to view all logs.
2. The system displays all activity logs sorted newest to oldest.		If no logs exist for the selected date, the system displays: "No records found."
3. Caregiver optionally applies a date filter.		Skip filtering to view all logs.
4. The system displays filtered log entries including reminder and recognition events.		
Post Condition:	Caregiver successfully reviews patient activity history including completed, missed reminders, and recognition events.	

2.4.12 UCD012 – Manage Patient Profile

Use Case ID: 012	**Use Case Name: Manage Patient Profile**
Priority: Medium	
Actors: Caregiver	

Use Case Summary:	The caregiver edits an existing patient profile to update name, age, relationship, or photo. Changes are saved and reflected across all linked data.
Pre-Condition:	Caregivers must be logged in and a patient profile must already exist.

Normal Flow of Events	**Alternative Path**
1. Caregiver navigates to the patient profile from the Dashboard.	Press cancel to discard changes.
2. Caregiver clicks "Edit Profile."	If mandatory fields are empty, the system shows a validation error.
3. Caregiver updates desired fields: Name, Age, Relationship, or Photo.	
4. Caregiver clicks "Save Changes."	
5. System validates and updates the profile in Firebase Firestore.	
6. Updated profile is reflected across the Dashboard.	

Post Condition:	The patient profile is successfully updated. All linked data remains associated with the same Patient ID.

2.4.13 UCD013 – Toggle Auto-Recognition Mode

Use Case ID: 013	**Use Case Name: Toggle Auto-Recognition Mode**
Priority: High	
Actors: Caregiver	
Use Case Summary:	The caregiver enables or disables Automatic Face Recognition Mode from Settings. When disabled, face recognition can only be triggered manually by the patient.
Pre-Condition:	Caregivers must be logged in and the patient device must be paired.

Normal Flow of Events	Alternative Path
1. Caregiver navigates to Settings from the Dashboard.	Press cancel to discard.
2. Caregiver locates the "Auto-Recognition Mode" toggle.	
3. Caregiver enables or disables the toggle.	
4. Caregiver clicks "Save."	
5. System updates the setting for the linked Patient ID.	
6. The patient's device reflects the updated mode immediately.	
Post Condition:	Auto-Recognition Mode is toggled. If disabled, recognition activates only manually.

2.4.14 UCD014 – Switch Between Patients

Use Case ID: 014	**Use Case Name: Switch Between Patients**
Priority: Medium	
Actors: Caregiver	
Use Case Summary:	A caregiver managing multiple patients can switch between profiles from the Dashboard. Each patient's data remains fully isolated.
Pre-Condition:	Caregivers must be logged in with more than one patient profile created.
Normal Flow of Events	**Alternative Path**
1. Caregiver accesses the patient selector from the Dashboard.	
2. The system displays a list of all linked patient profiles.	
3. Caregiver selects the desired patient profile.	

4. Dashboard updates to display that patient's data.	
Post Condition:	Dashboard reflects the selected patient's reminders, logs, and visitor data. All other patient data remains isolated.

2.4.15 UCD015 – Caregiver Logout

Use Case ID: 015	**Use Case Name: Caregiver Logout**
Priority: High	
Actors: Caregiver	
Use Case Summary:	The caregiver logs out to secure their account. The session is terminated and the user is redirected to the Login screen.
Pre-Condition:	Caregivers must be currently logged into the system.
Normal Flow of Events	**Alternative Path**
1. Caregiver clicks on the account menu or settings icon.	
2. Caregiver selects the "Logout" option.	Click elsewhere to cancel.
3. The system terminates the active session.	
4. Caregiver is redirected to the Login screen.	
Post Condition:	Caregiver session is successfully terminated. Application returns to the Login screen.

2.5 Use Case Diagram

2.5.1 Caregiver Diagram:

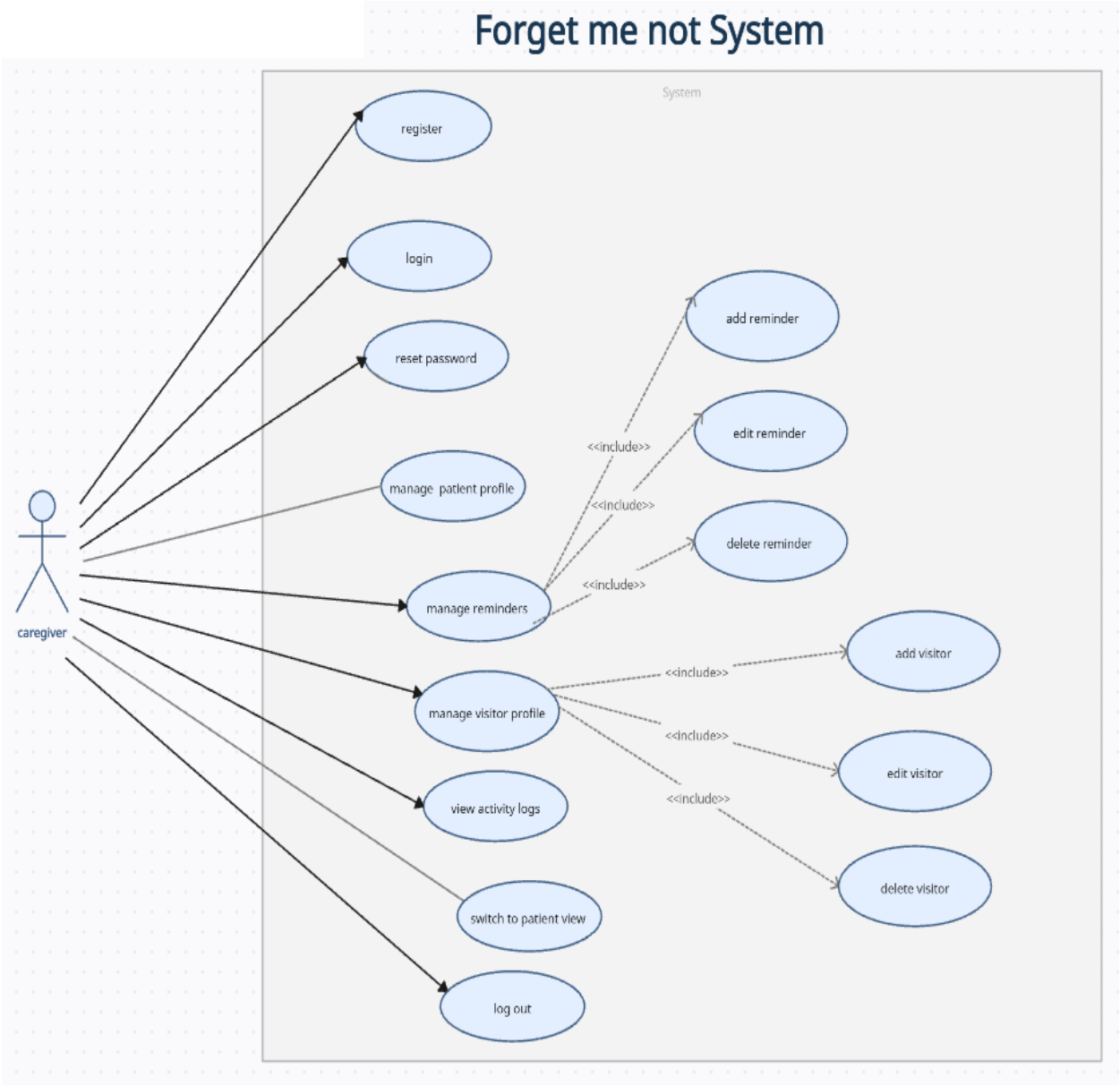


Fig 1 – UCD01: Caregiver

2.5.2 Patient Diagram:

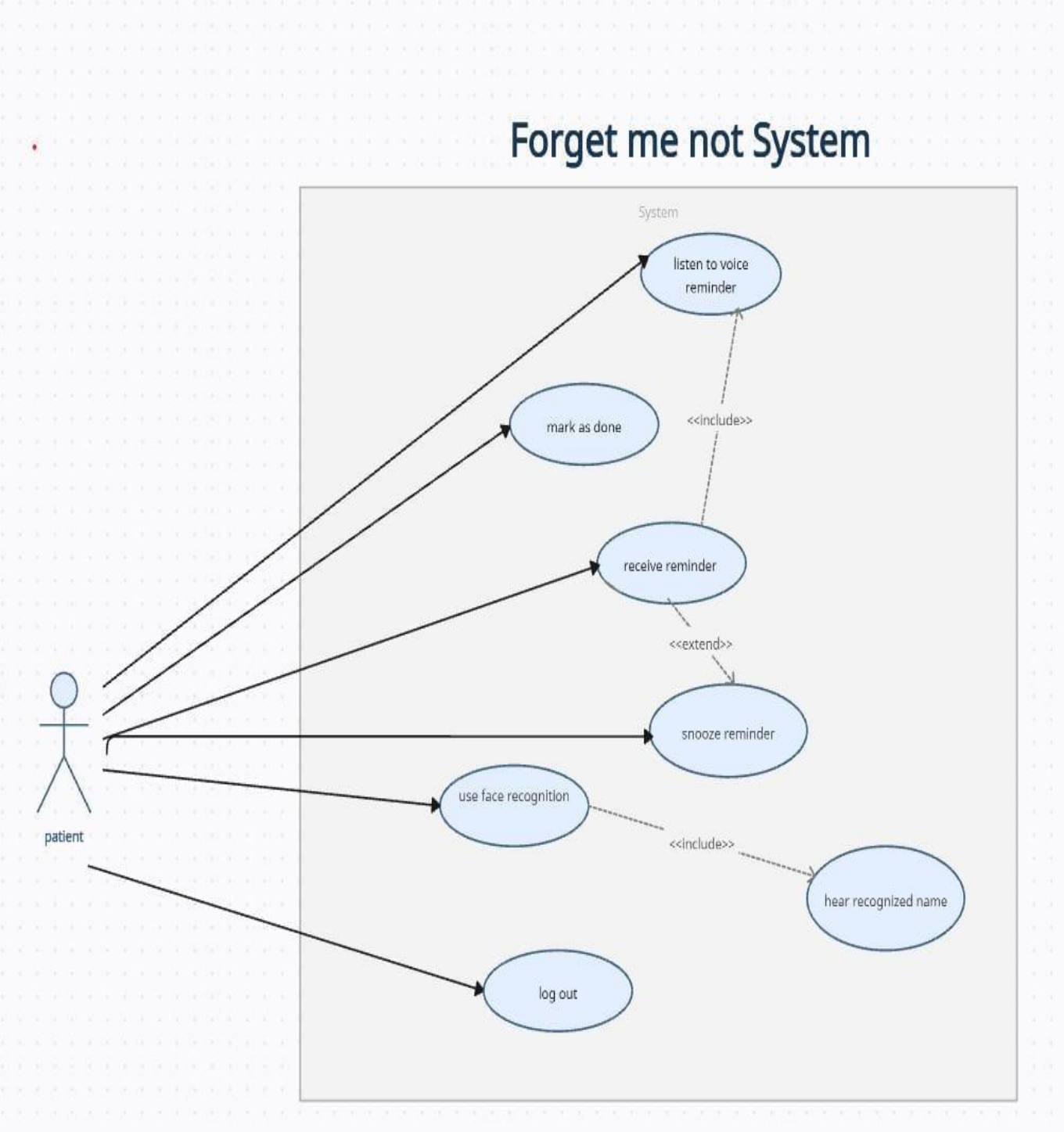


Fig 2 – UCD01: Patient

2.6 Software Development Life Cycle model (Justification on why this particular model is considered)

SDLC has different type of models for example V-model, Water Fall model, iterative model etc. the purpose of these models is to explain the phases of your product development for very first step to the last step.

The SDLC model that we are going to focus on is Incremental model because in this type of model requirements are divided into groups at the start of the project and it is similar to Waterfall model. For each group, the SDLC model is followed waterfall cycle to develop software. The SDLC process is repeated, with each release adding more functionalities until all requirements are met. As in this semester we have analysed our problem and gathered all the information and further testing and implementation is going to be done in next semester.

We will be working on implementation of modules for students first and also we will continuously improve design while implementing and add further features, so we are choosing Incremental model.

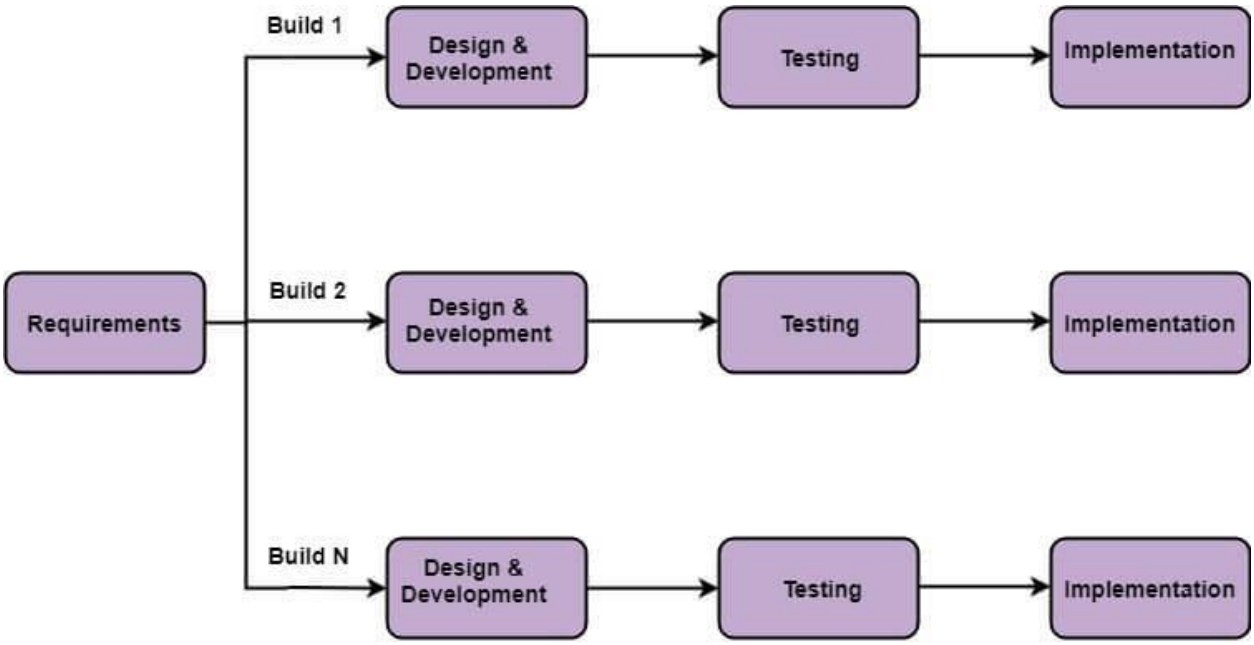


Fig 3 – SDLC01: Incremental model

CHAPTER 3

3 System Design

3.1 Work Breakdown Structure (WBS)

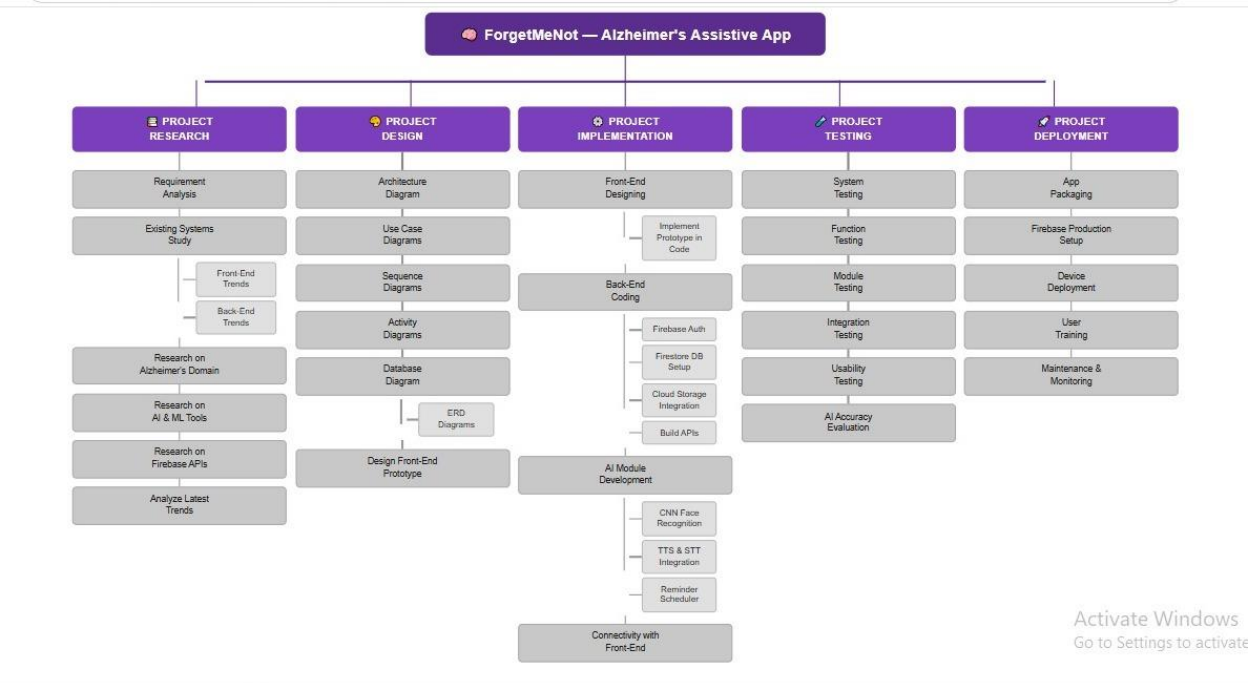


Fig 4 – WBS01: ForgetMeNot–Alzheimer’s Assistive App

3.2 Activity Diagram

3.2.1 AD001- Caregiver Registration Screen

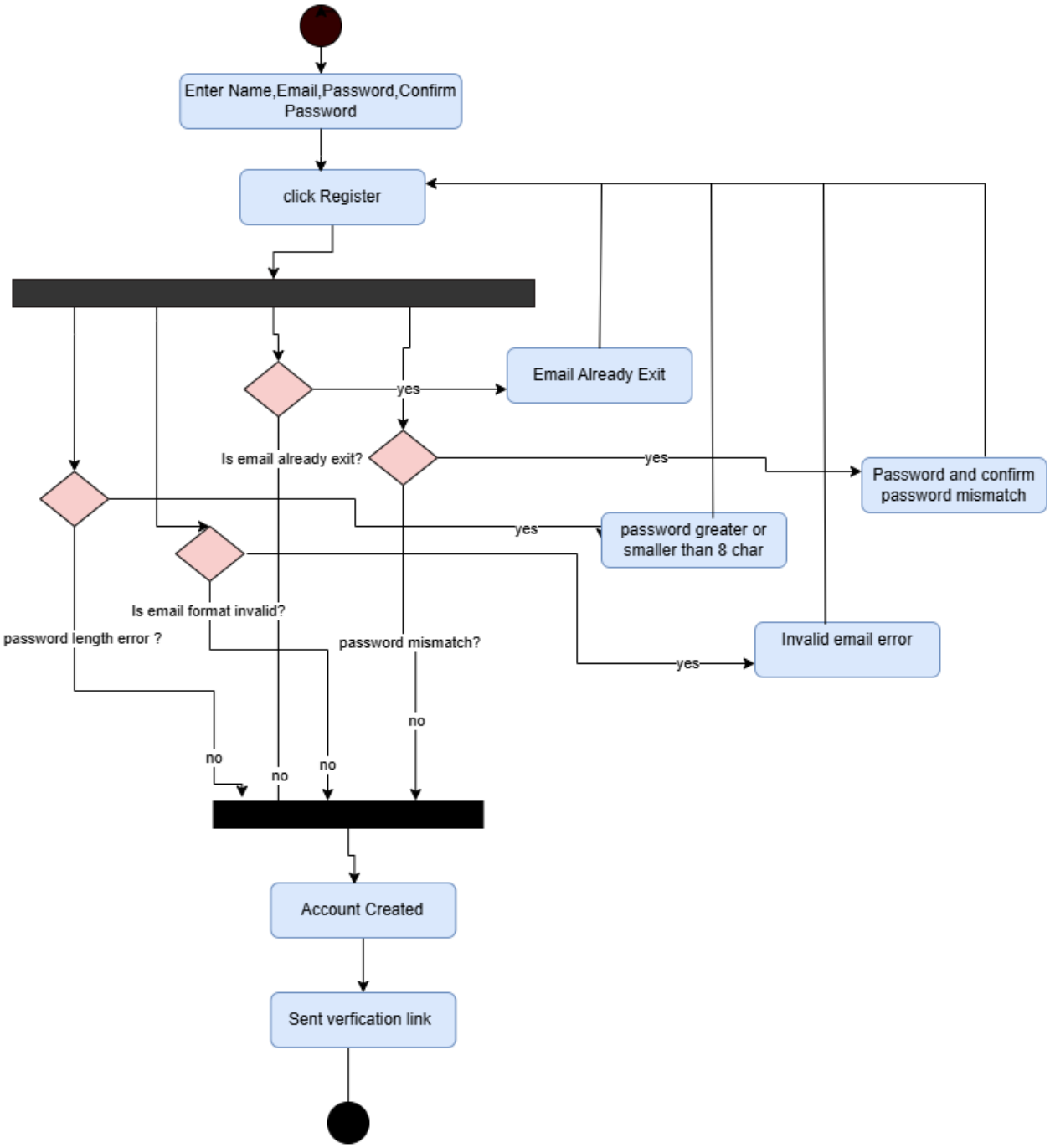


Fig 5 –AD001: **Caregiver Registration Screen**

3.2.2 AD002-Login

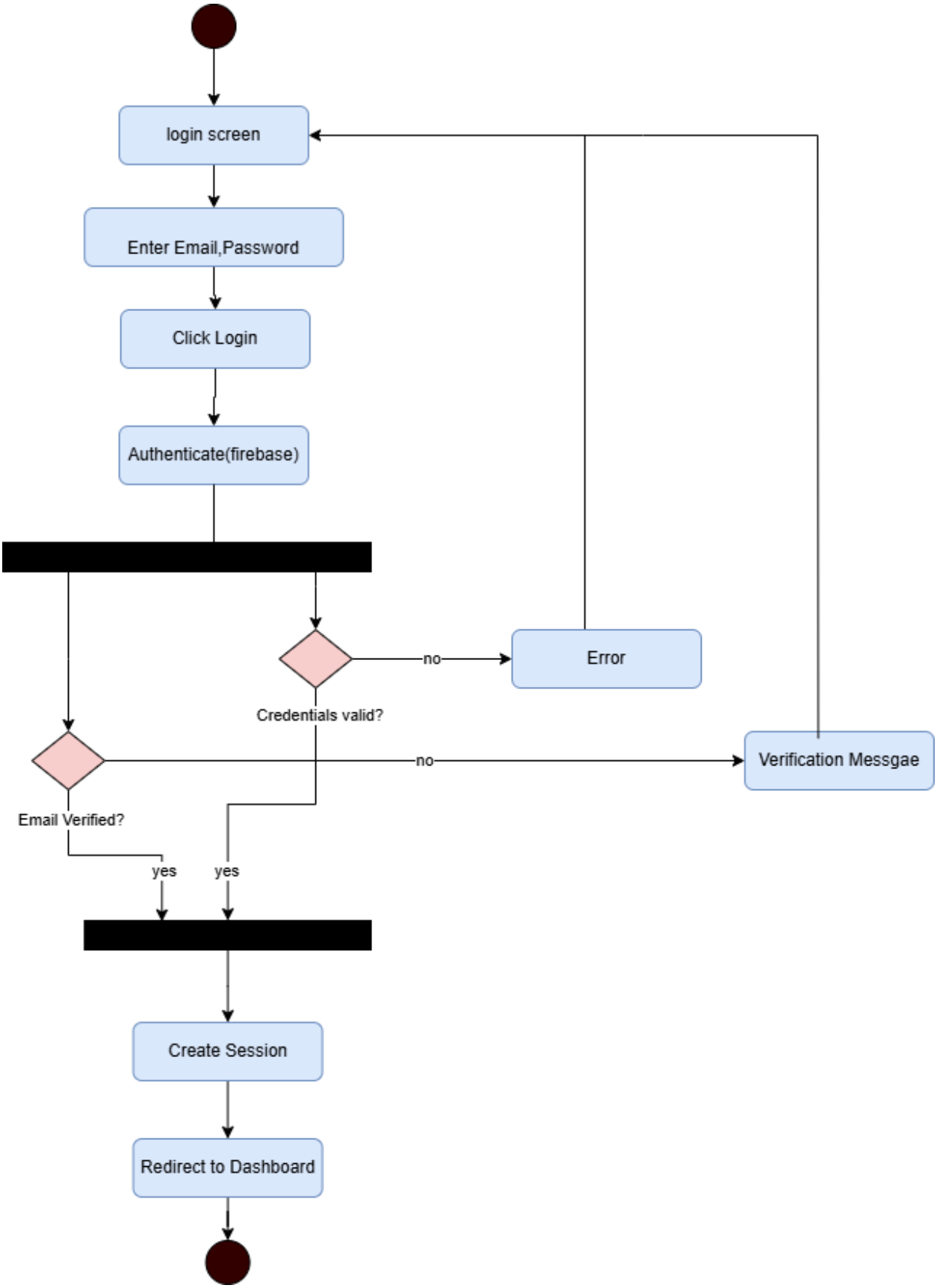

login screen
Enter Email,Password
Click Login
Authenticate(firebase)
Credentials valid?
no
Error
Email Verified?
no
Verification Messgae
yes
yes
Create Session
Redirect to Dashboard

Fig 6 –AD002: Login

3.2.3.1 Initial Patient Setup

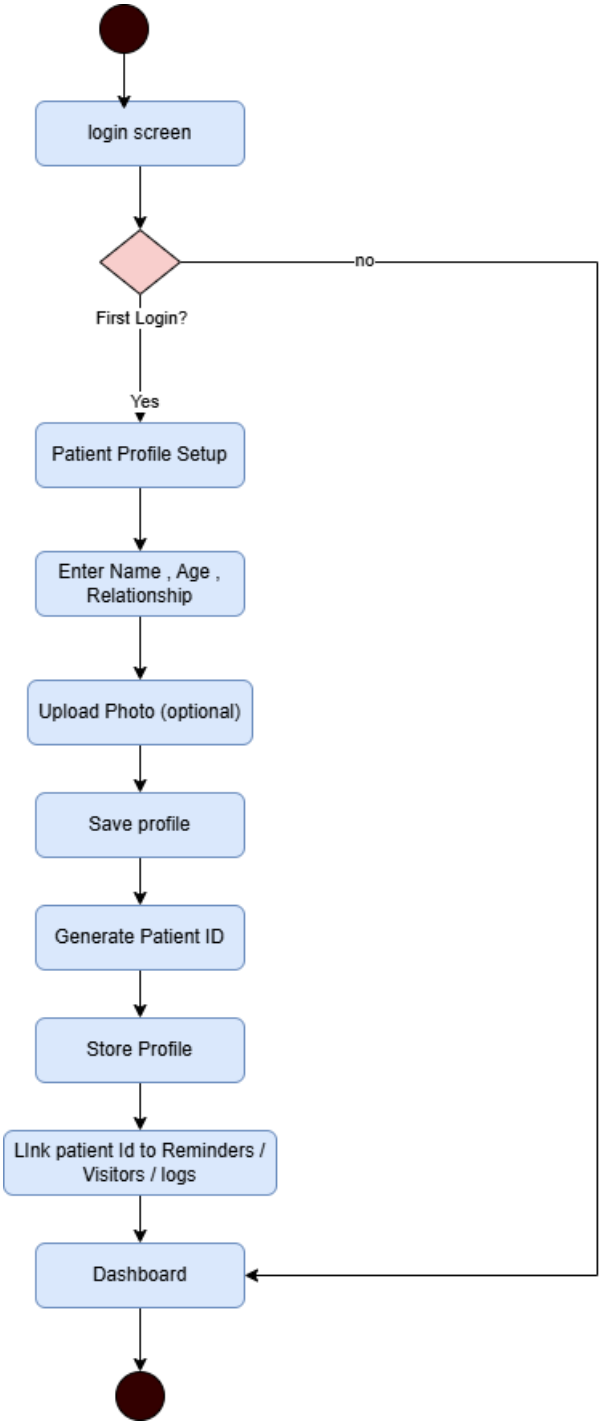


Fig 7 –AD003: **Initial Patient Setup**

3.2.3.2 Device Pairing (Patient Mode Activation)

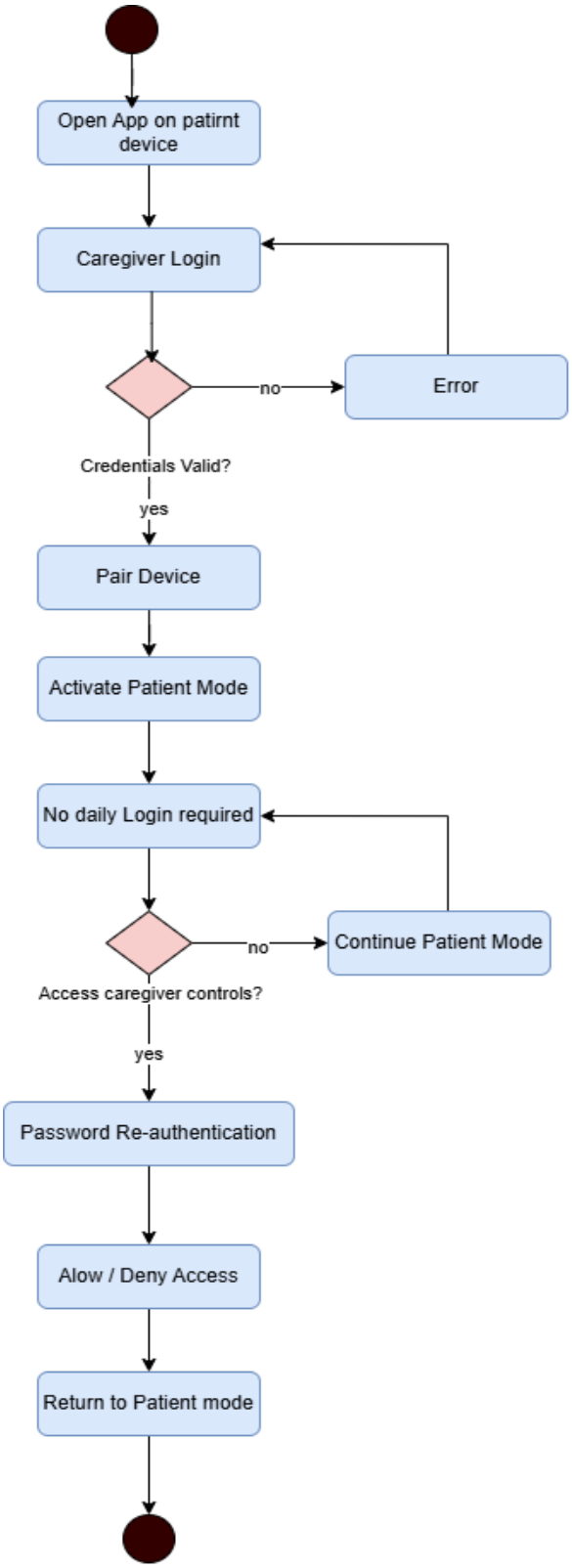
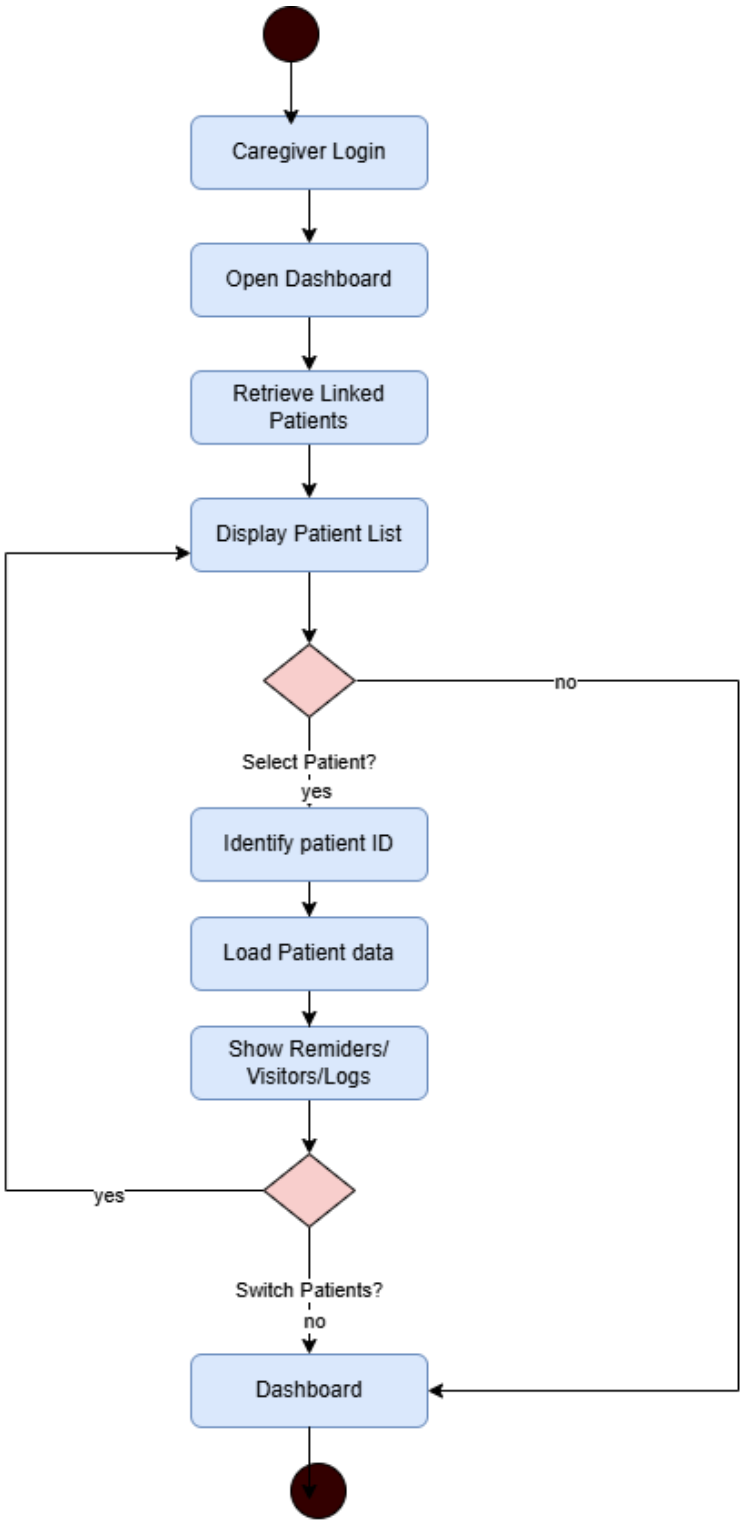
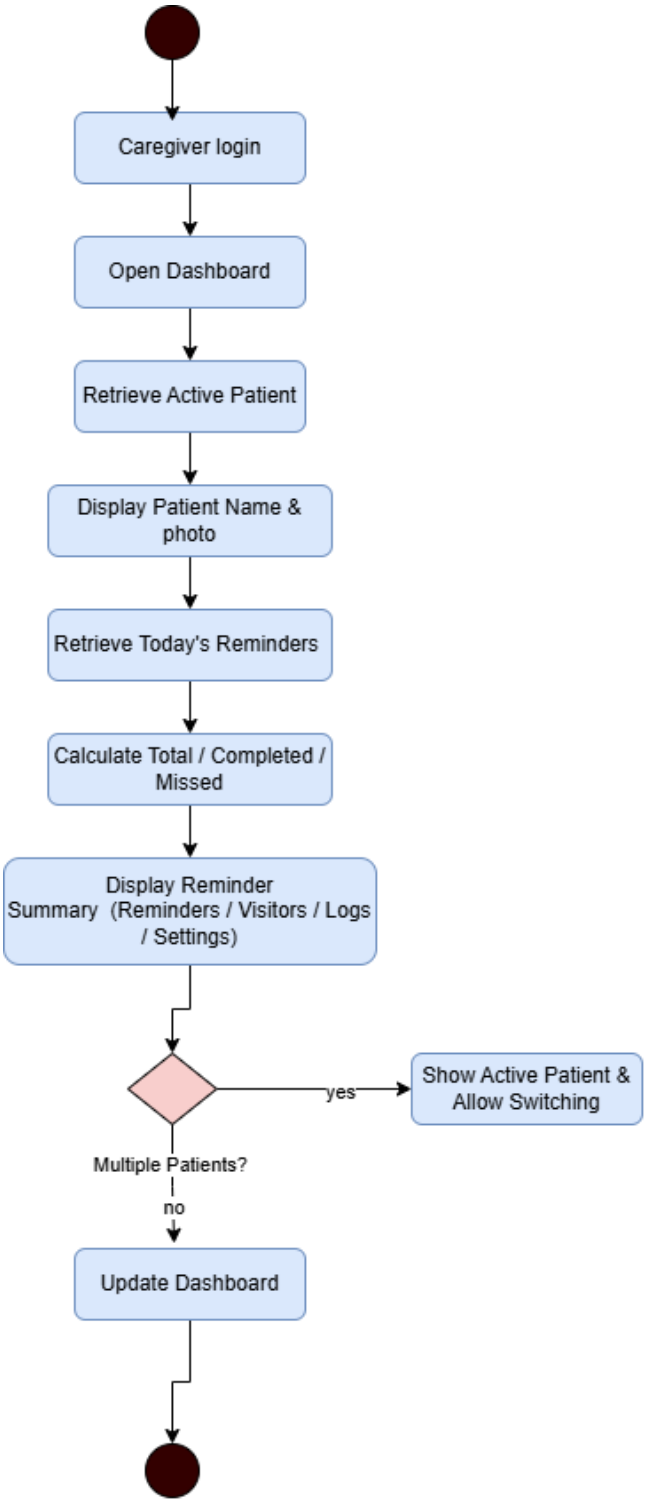


Fig 8 –AD004: **Device Pairing**

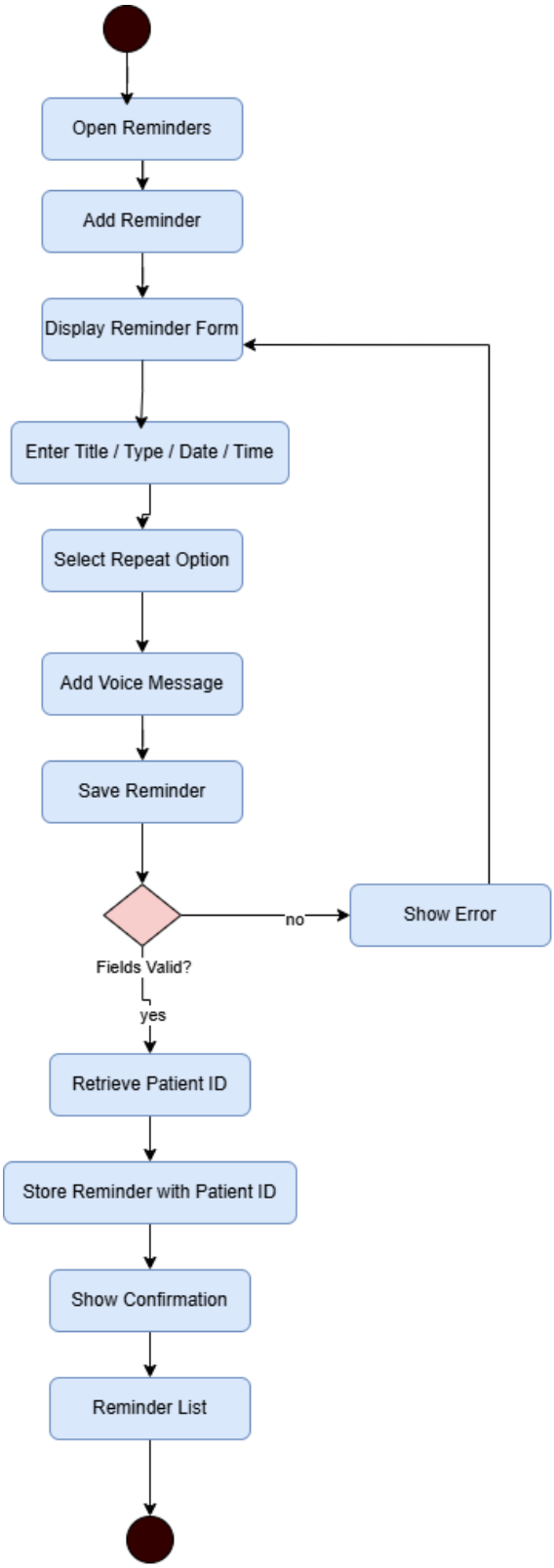
3.2.3.3 Multiple Patient Support (Secondary Feature)



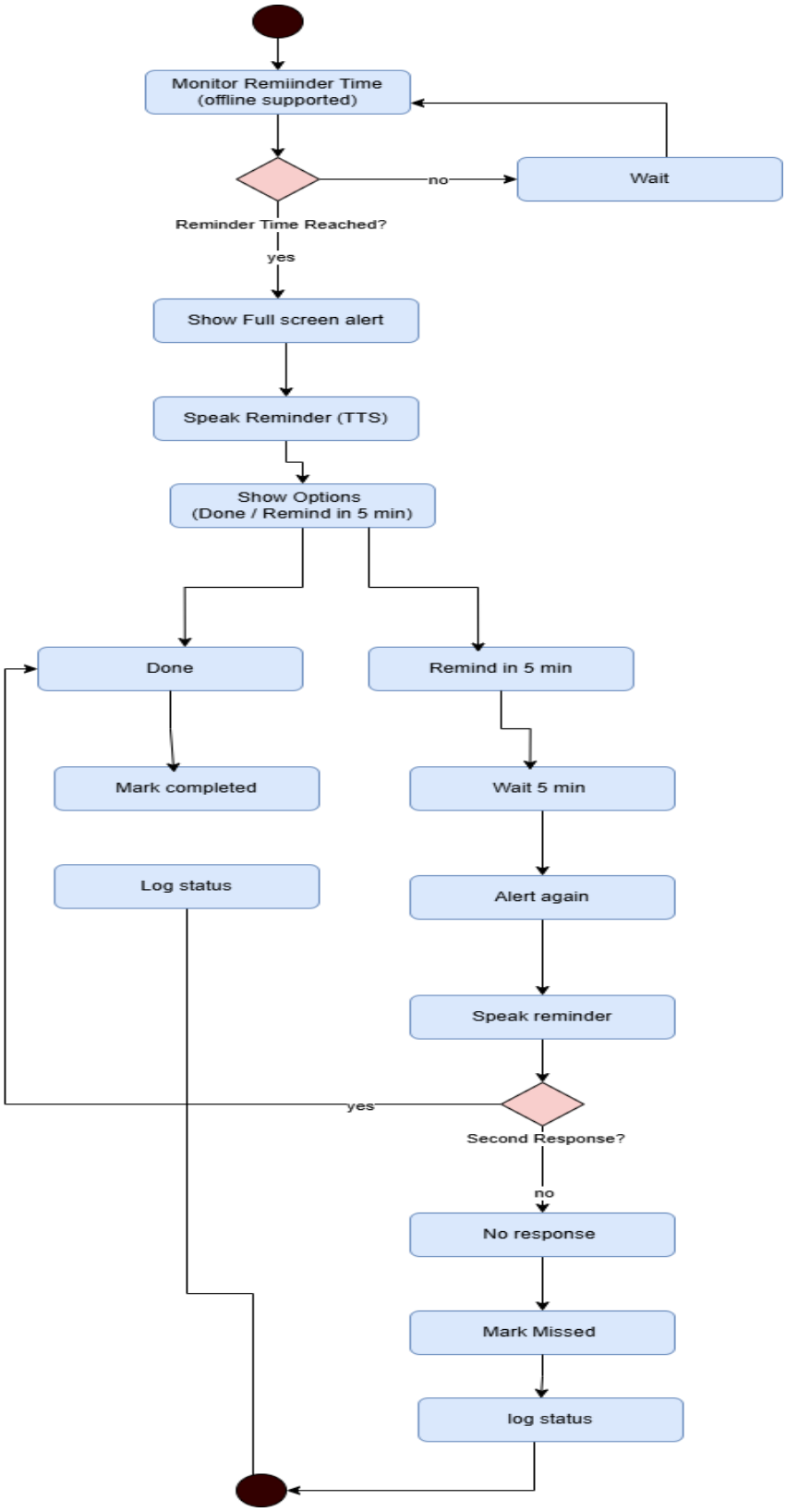
3.2.4. Caregiver Dashboard



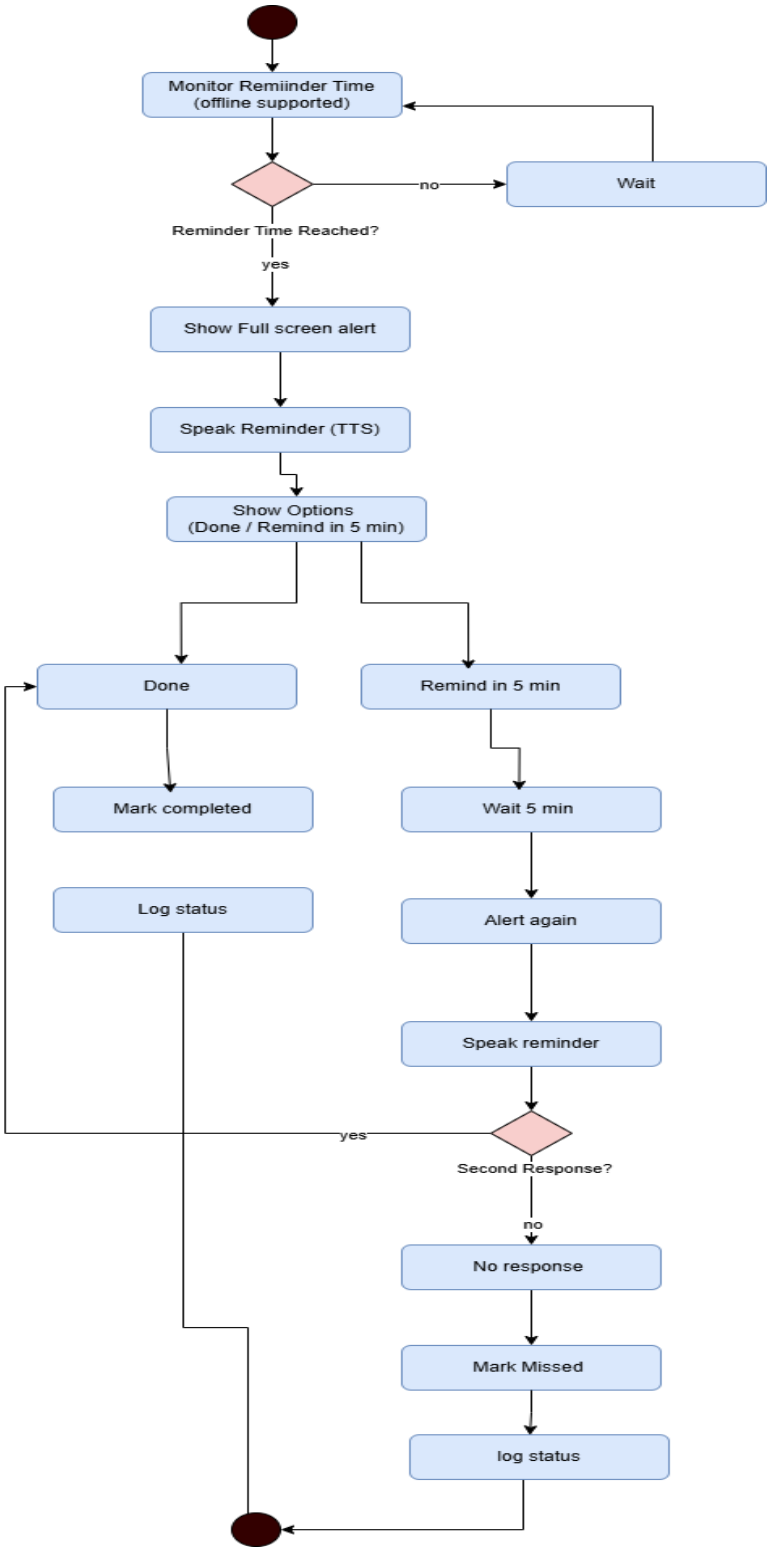
3.2.5.1 Add Reminder (Caregiver)



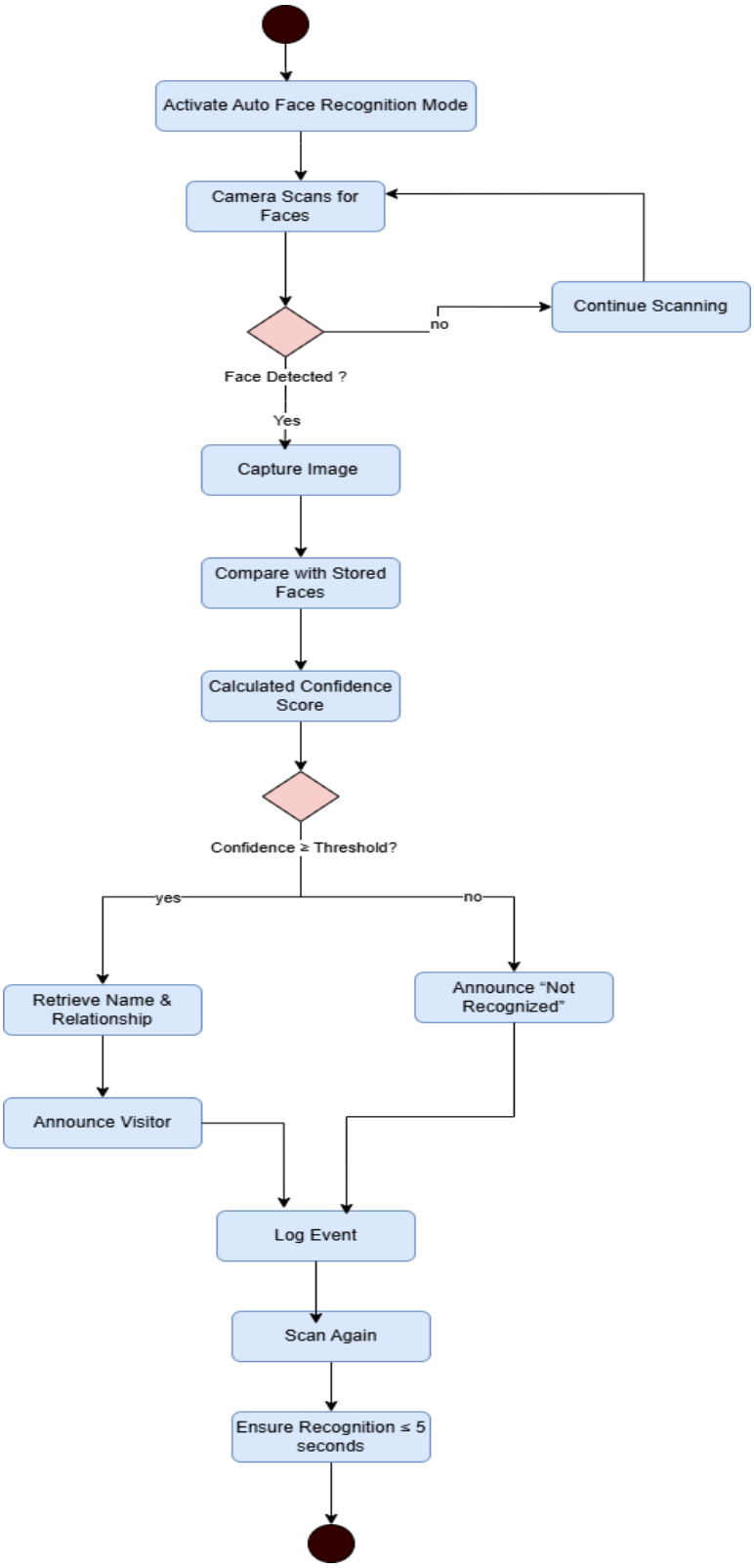
3.2.5.2 Reminder Execution (Patient Side)



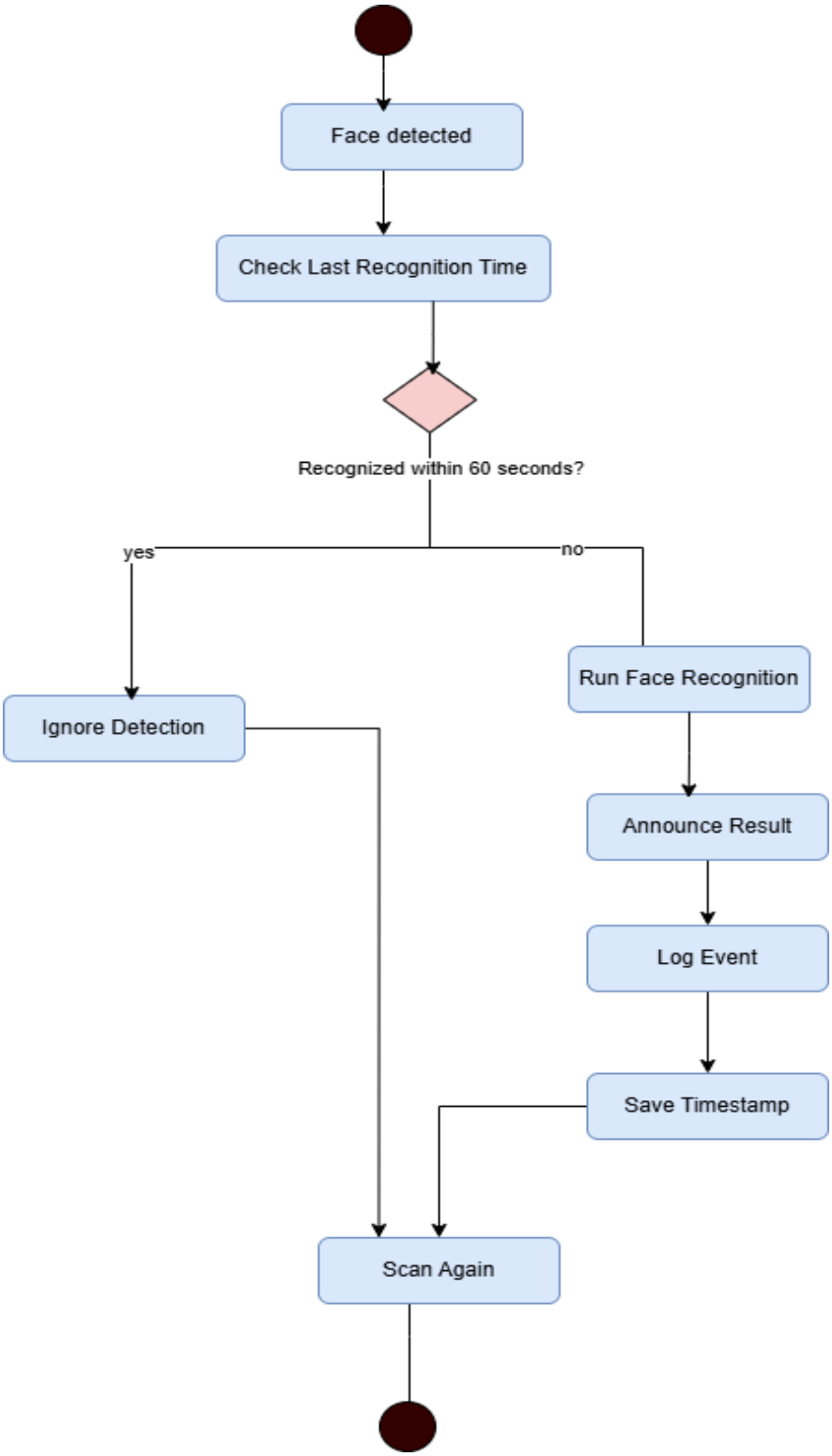
3.2.6 Visitor Management (caregiver)



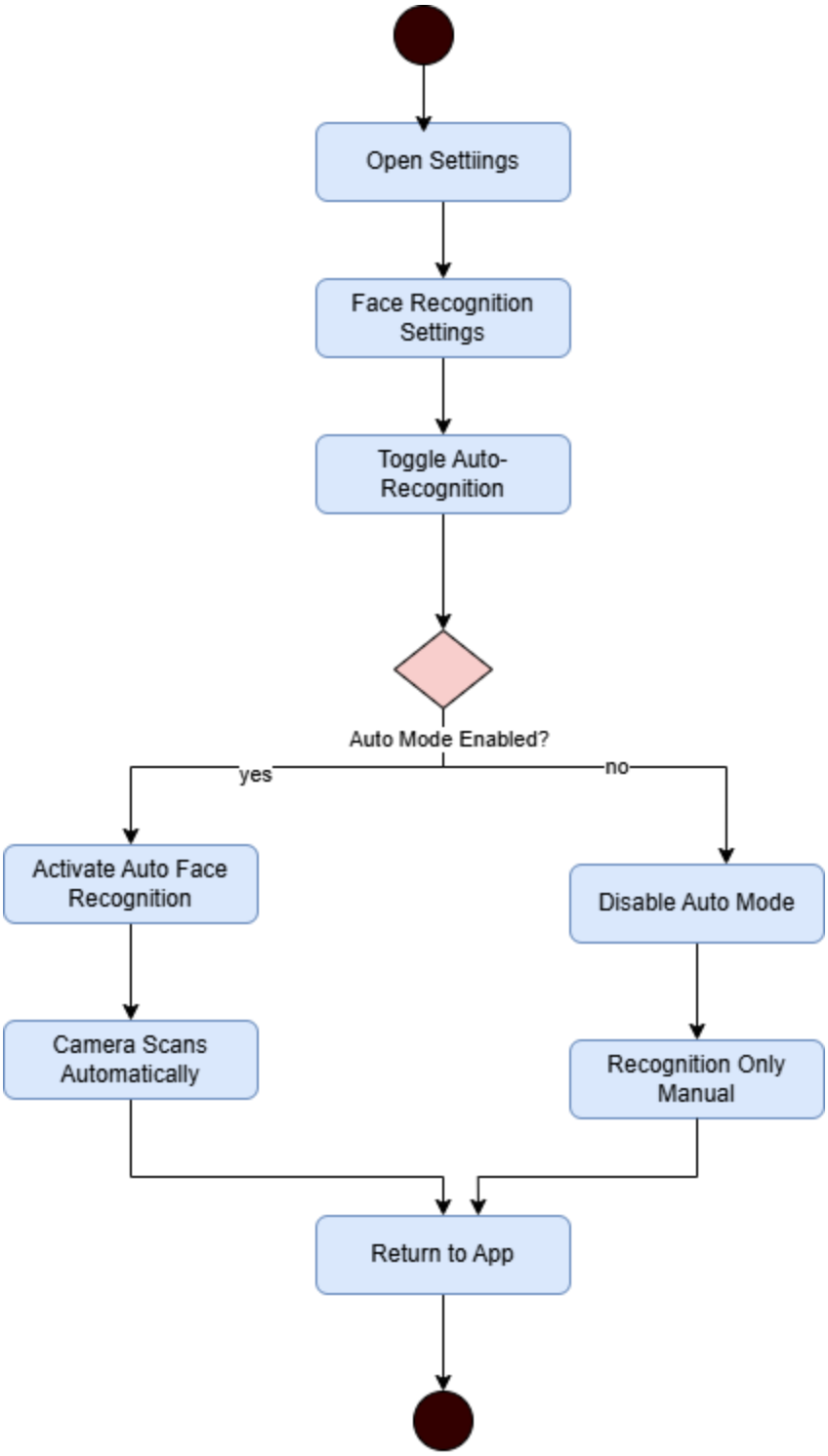
3.2.7.1 Auto-Recognition Logic



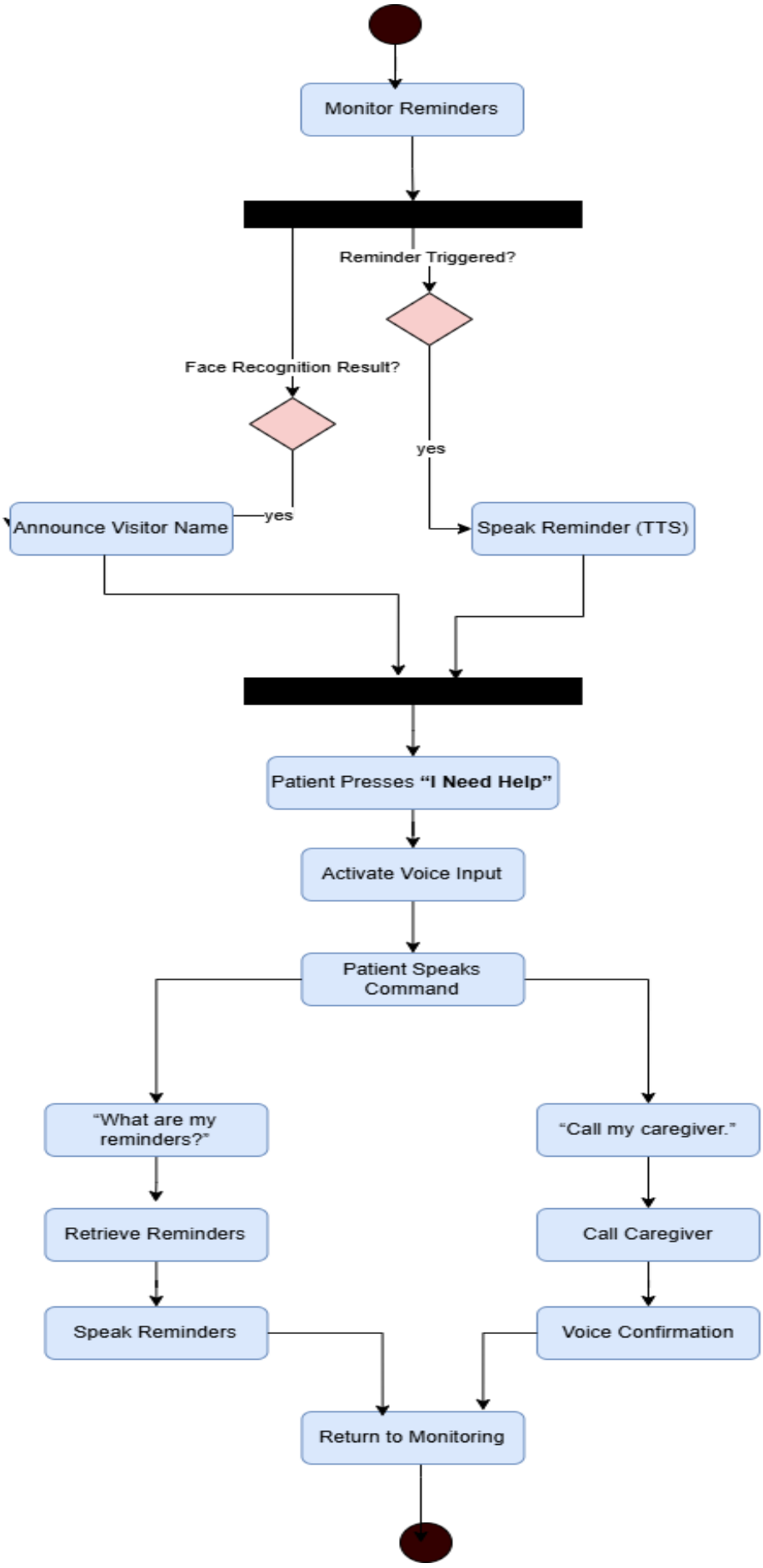
3.2.7.2 Cooldown Mechanism



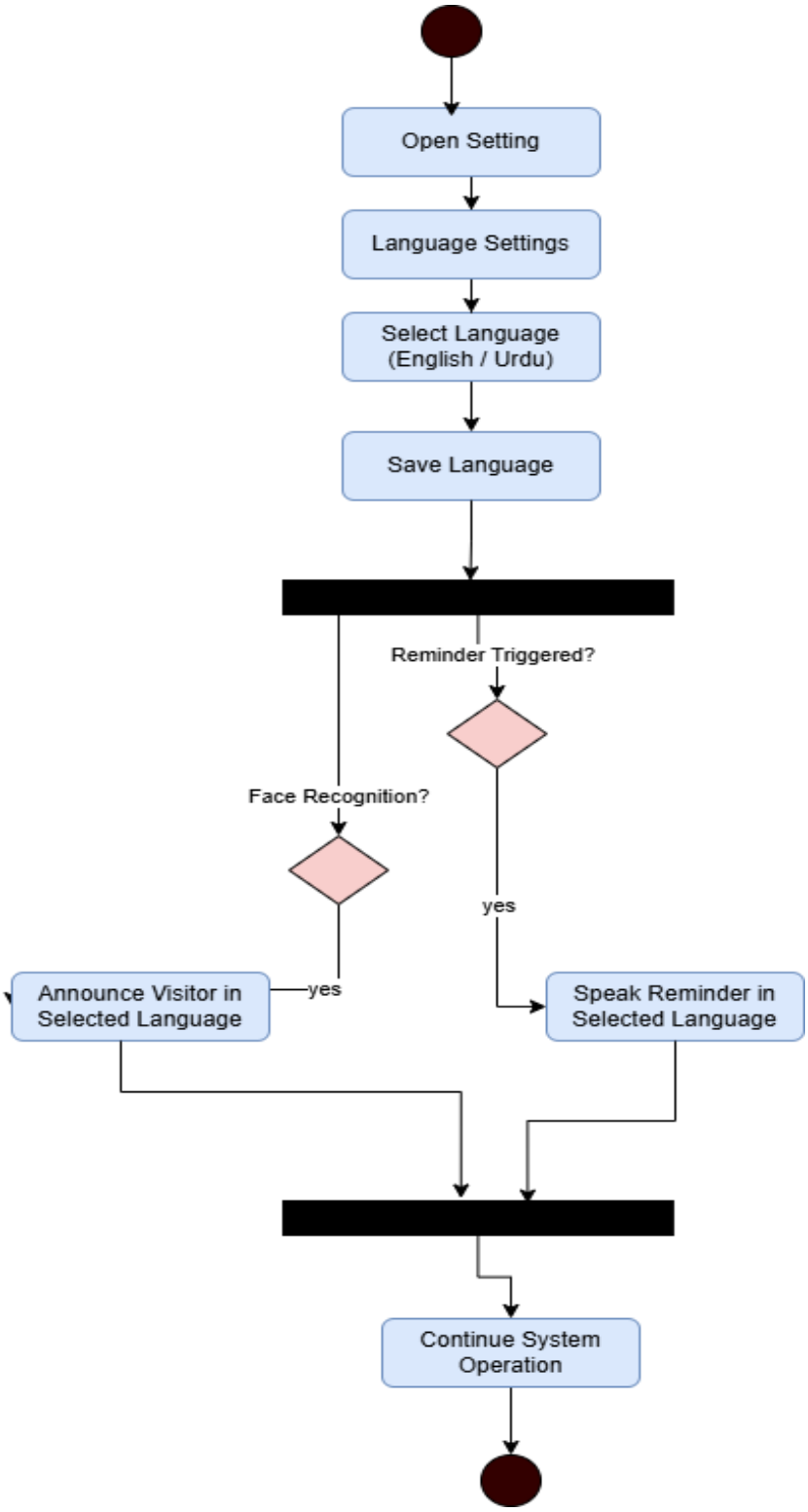
3.2.7.3 Auto-Recognition Control (caregiver)



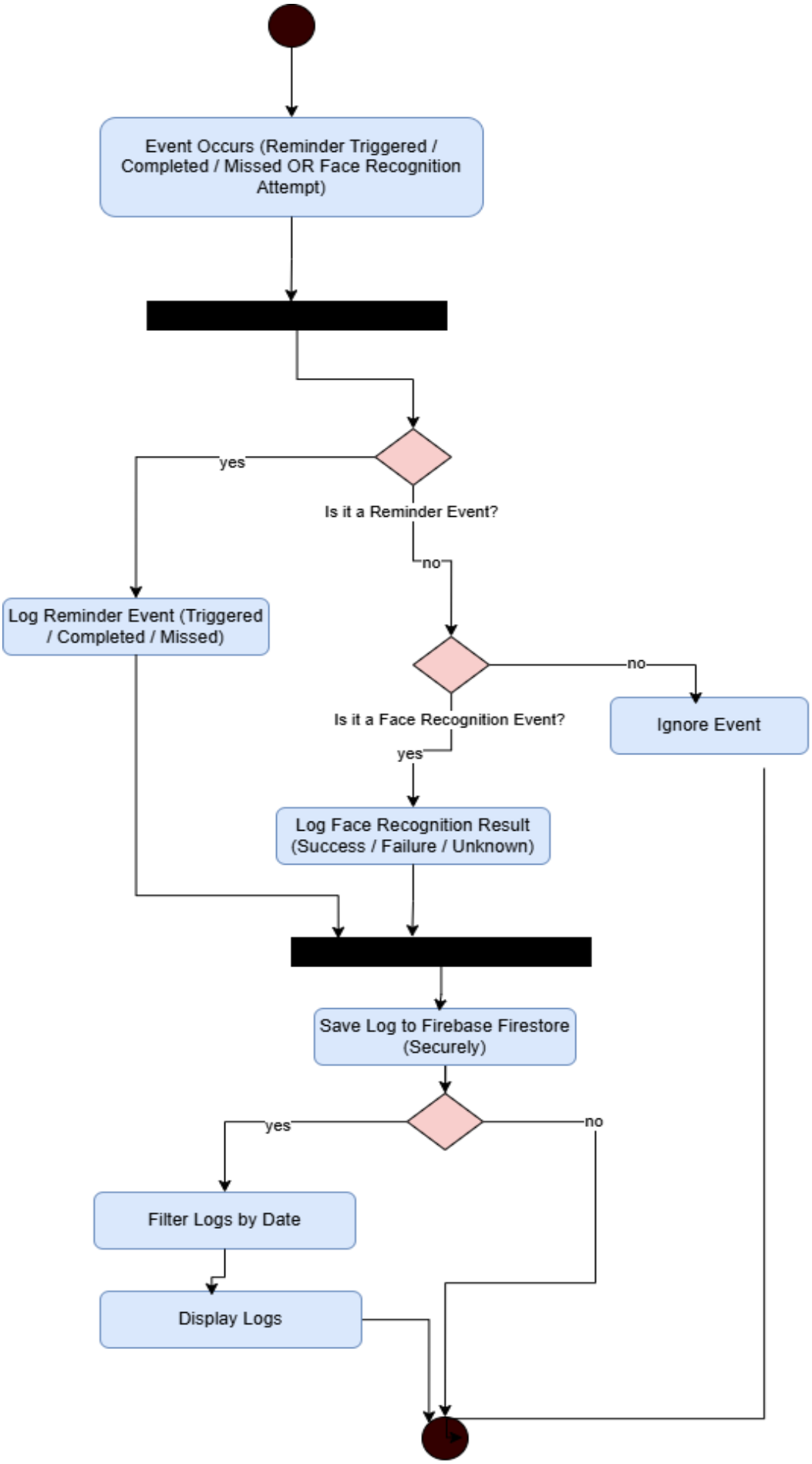
3.2.8. Voice Assistant


Monitor Reminders
Reminder Triggered?
Face Recognition Result?
yes
yes
Announce Visitor Name
Speak Reminder (TTS)
Patient Presses "I Need Help"
Activate Voice Input
Patient Speaks Command
"What are my reminders?"
"Call my caregiver."
Retrieve Reminders
Call Caregiver
Speak Reminders
Voice Confirmation
Return to Monitoring

3.2.9. Language Selection



3.2.10. Activity Logs


Event Occurs (Reminder Triggered / Completed / Missed OR Face Recognition Attempt)
yes
Is it a Reminder Event?
no
Log Reminder Event (Triggered / Completed / Missed)
no
Ignore Event
Is it a Face Recognition Event?
yes
Log Face Recognition Result (Success / Failure / Unknown)
Save Log to Firebase Firestore (Securely)
yes
no
Filter Logs by Date
Display Logs

7 References

[1] C. Fang et al., “Impaired holistic processing of facial recognition in patients with Alzheimer’s disease,” *Neurology*, vol. 94, no. 15_Supplement, 2020.

[2] U. Akare, P. Lonkar, A. Rangari, R. Bhandakkar, and R. Gaikwad, “Research on Assistive System for Alzheimer Patient,” *International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE)*, vol. 14, no. 4, Apr. 2025.

[3] A. Paul, “Wearable Facial Recognition Device for Dementia,” Regeneron ISEF 2023 Project ENBM084.

[4] B. Peres and P. F. Campos, “A systematic review of reminder and guidance systems for Alzheimer’s Disease and Related Dementias: context, barriers and facilitators,” *Disability and Rehabilitation: Assistive Technology*, 2023.

[5] MIT Media Lab, “Memoro: Wearable Memory Assistant using Personal AI,” ArXiv preprint, 2024.

[6] RAZ Mobility, “RAZ Memory Cell Phone — The Simple Phone for Dementia, Alzheimer’s, And Seniors,” Product Website, 2024.